AI 101 for Mechanical Engineers · Module 6

Parameters and hyperparameters

Lesson 6.2

What you will learn

> Distinguish learned parameters, hyperparameters and supplied engineering assumptions, then separate the training loss from engineering acceptance criteria.

Lesson overview

Module 6 · Lesson 2

Parameters and hyperparameters

AI 101 for Mechanical Engineers

1 Identify how each value is obtained

Learned parameters	**Values obtained during fitting.** Examples: regression coefficients, neural weights and biases, tree split values.
Hyperparameters	**Settings that control model structure or training.** Examples: maximum tree depth, learning rate, batch size. Usually chosen using validation data.
Supplied physical values	**A fixed pump efficiency may be an engineering assumption.** It is a learned ML parameter only if fitting estimates it from data.

A fitted slope and a chosen depth limit have different roles.

2 Loss and acceptance answer different questions

Training loss	Engineering acceptance criteria
Guides fitting using a chosen objective.	State the requirements that assessed performance must meet.

MAE and MSE emphasize errors differently

Mean absolute error (MAE)	Averages the sizes of residuals, ignoring their signs.
Mean squared error (MSE)	Squares residuals before averaging, so larger errors have more influence.

A low training loss does not establish safe engineering use. An average can hide rare large failures. Inspect errors in engineering units and check acceptance requirements separately.

Key takeaway **Separate fitted values, training settings and physical assumptions. Evaluate acceptance separately.**

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Parameters are values obtained within fitting

- Examples: regression coefficients, neural weights and biases, tree split values
- Role follows how a value is obtained and used
- Not simply whether it appears as a number in software
- Fitted intercept and slope stay fixed during routine inference

Hyperparameters govern model structure or training

- Settings that govern model structure or training [4]
- Examples: maximum tree depth, learning rate, batch size
- Usually selected through validation, not fitted with the parameters
- The distinction depends on the method and value’s acquisition

Three kinds of value play different roles

Value	Role	Example
Learned parameter	Obtained within model fitting	Fitted regression slope
Hyperparameter	Governs permitted structure or training	Chosen maximum tree depth
Supplied engineering parameter	Model assumption for a calculation	Fixed pump efficiency

> A supplied value is a learned ML parameter only if estimated during fitting.

Training loss and acceptance criteria differ

Training loss

- Disagreement with the chosen training objective
- Example: mean squared error
- Guides fitting; not a universal quality or safety score

Metrics and acceptance

- Evaluation metrics describe assessed performance
- Acceptance criteria state required performance
- Both can differ from the training objective

An aggregate error can conceal rare large failures; lower loss does not prove every condition acceptable.

MAE and MSE emphasize different errors

Mean absolute error

- Averages absolute residual magnitudes [35]

Mean squared error

- Squares residuals before averaging [35]
- Gives large residuals more influence

Neither encodes asymmetric consequences or defines an acceptance limit. Inspect errors in engineering units.

CHECK YOUR UNDERSTANDING

Which is a hyperparameter: a fitted slope or a chosen maximum tree depth?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Which is a hyperparameter: a fitted slope or a chosen maximum tree depth?

> The chosen maximum tree depth is a hyperparameter. The slope obtained during fitting is a parameter.

CHECK YOUR UNDERSTANDING

Does a low training loss establish safe engineering use?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a low training loss establish safe engineering use?

> No. Engineering acceptance needs its own criteria and supporting evaluation; the training objective alone does not establish safety.

Key takeaways

- Learned parameters and chosen training settings serve different roles
- Supplied physical assumptions are not automatically learned ML parameters
- State the chosen loss and its purpose
- Evaluate engineering acceptance separately from training loss

References

[4] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

[35] C. J. Willmott and K. Matsuura, “Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance,” Clim. Res., vol. 30, pp. 79–82, 2005. doi: 10.3354/cr030079

Numbers match the course reference list. The written lesson is the authoritative text.